

Cloud Based Implementation of IoT Using MQTT Brokers

Message Queuing Telemetry Transport (MQTT) is a standardised, lightweight protocol that transmits messages between connected objects in an IoT network.

A number of devices and gadgets communicate in real-time using advanced wireless technologies. Smart cities, smart offices, smart homes, etc, are interconnected specifically by the IoT in which gadgets and smart objects communicate with each other using specific wireless protocols. IoT is also referred to as Ubiquitous Computing or Pervasive Computing, and there are many real-time applications of these technologies. The FASTags introduced in the country recently to pass through toll plazas is a classical implementation of IoT, in which a small chip on a vehicle communicates with the nearby sensor and a transaction is done automatically.

In such scenarios, the scalability as well as reliability issues are addressed by high-performance network communications or advanced wireless communications using 4G, 5G and related technologies.

Protocols and engines for the implementation of Internet of Things

IoT helps smart gadgets communicate using less resources and minimum power consumption. In the IoT environment, MQTT is a specific protocol for low-powered and lightweight communication so that there is minimum load on devices and gadgets. It is the key data protocol used in IoT for swift communication in low-powered devices.

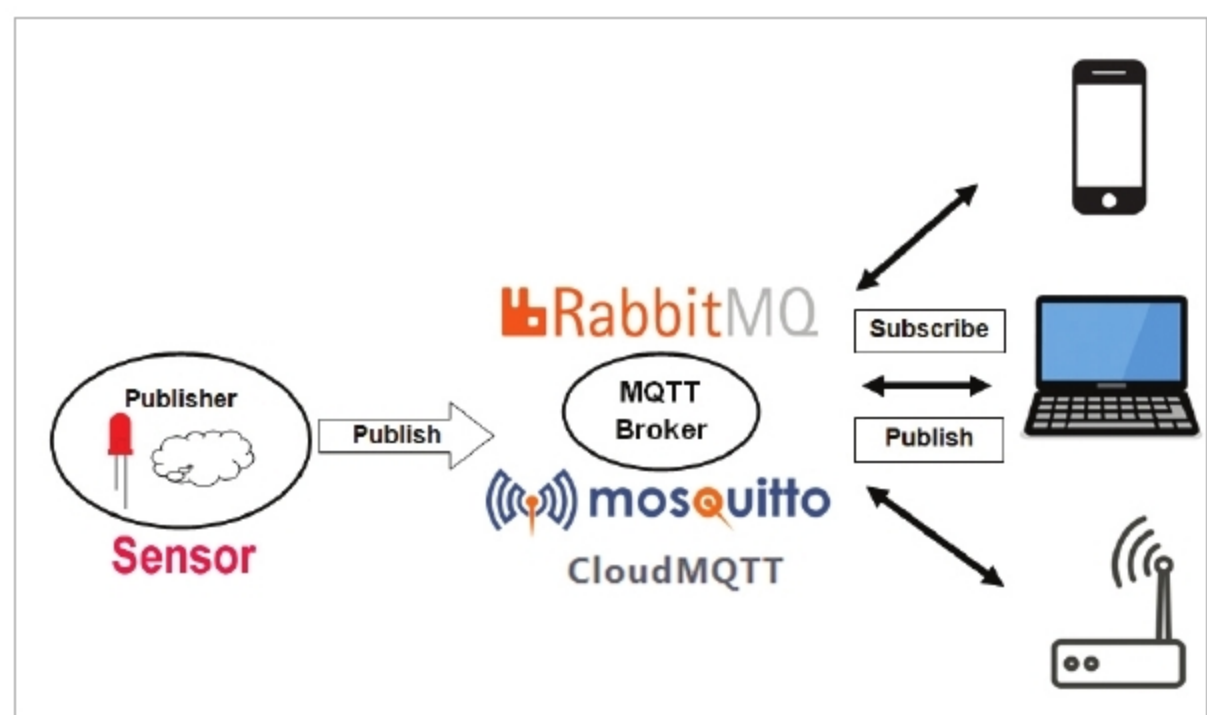


Figure 1: Key constituents of MQTT

Lightweight and low-powered communication needs to be speedy and reliable in order to operate on high frequency network communications with various devices. MQTT enables this communication between a large number of devices and gadgets with varying configurations.

Dimensions of MQTT broker

In IoT communication, there are mainly two types of objects — publisher and subscriber. Data signals transmission is known as publishing. Technically, the transfer of data from the computer to a separate end is called publishing.

Layer	Protocols
Data protocols	MQTT, AMQP, CoAP, Node, Websocket
Infrastructure	6LowPAN, RPL, IPv4/IPv6
Communication transport	Wi-Fi, LPWAN, Bluetooth
Semantics	Web Thing, JSON-LD
Discovery	Physical Web, DNS-SD, mDNS
Device management	OMA-DM, TR-069
Identification	URLs, EPC, IPv6, uCode

Table 1: Key protocols in IoT communication

